
Certificate

Dr. Thejus Mahajan

has successfully completed an advanced vocational training

Applied Bioinformatics and Biostatistics

from 08/18/2025 to 04/16/2026

at

CQ Beratung+Bildung GmbH
Strelitzer Straße 60, 10115 Berlin, Germany

Contents

◆ Introduction in Programming and Python	(224 TU*)
◆ Bioinformatics and Databases	(120 TU)
◆ Applied Bioinformatics	(120 TU)
◆ Applied Biostatistics	(256 TU)
◆ Next-Generation Sequencing	(80 TU)
◆ Internship	(11 weeks)

Berlin, 04/16/2026



Management



* One Teaching Unit (TU) corresponds to 45 min.

** It applies to grades by IHK Berlin.ie

Certificate

Applied Bioinformatics and Biostatistics

Dr. Thejus Mahajan

Contents of training	Grade**
1. Introduction in Programming and Python	1.2
Job application coaching, Job profile and perspectives, Shell scripting under LINUX, Programming methodology, Pseudocode, Data structures, Basics of programming with Python (including Numpy, Pandas), GUI (Graphical user interface) Design (Tkinter), Object-oriented programming with Python, Introduction to version control with Git	
2. Bioinformatics and Databases	1.4
(Molecular) Biological/Sequence/Interaction databases / NCBI, Biopython: Web Services, Database Management – SQL / Python	
3. Applied Bioinformatics	1.5
Sequence analysis, Pairwise/ multiple sequence alignment, BLAST, Dotplots, Structural bioinformatics and software for visualisation of structures (pymol), Digital image processing in Python	
4. Applied Biostatistics	1.0
Descriptive and multivariate statistics with R/R-Studio, Variance/principal component/cluster/discriminant analysis, R – Bioconductor, Implementation of Markov chains, Artificial neural networks, Machine learning in R	
5. Next-Generation Sequencing	2.1
Conda, NGS - Next generation sequencing in Nextflow, Variant Call, Mapping, RNA-Seq, Transcriptome Assembly, Demultiplexing	
6. Internship	
11 weeks at the HealthTwist GmbH in Berlin	

Berlin, 04/16/2026

Bilal Wier

Management



* One Teaching Unit (TU) corresponds to 45 min.

** It applies to grades by IHK Berlin.ie